UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FAKULTI TEKNOLOGI MAKLUMAT DAN KOMUNIKASI (FTMK)

(Under Student Mobility Program from Universitas Dian Nuswantoro, Indonesia)

A	**TITLE OF PROPOSED PROJECT :**
	Cross Dataset Validation of a LightGBM-Based Android Malware Detection Pipeline: A Generalizability Study Using the KronoDroid Dataset
B	**DETAILS OF STUDENT**
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C	**PROJECT INFORMATION**
C(i)	Android malware detection has become increasingly important as malicious applications become more sophisticated. A previous study developed a leakage-free detection pipeline that combines log1p transformation, RobustScaler normalization, and in-fold SMOTE oversampling, with LightGBM as the main classifier. The pipeline achieved 96.36% hold-out accuracy and 95.58% accuracy in 5-Fold Cross Validation on the CICMalDroid2020 dataset, outperforming previous benchmark methods by up to 3.08%. However, this pipeline was only evaluated on a single dataset, so its ability to generalize to other Android malware datasets has not yet been verified. This is an important limitation because a detection pipeline that performs well on one dataset may not perform equally well on data collected under different conditions, time periods, or feature extraction methods. This project proposes to evaluate the same pipeline on the KronoDroid dataset, specifically the real-device subset. KronoDroid also provides statistical features based on system calls, making it suitable for a methodologically consistent comparison. The study aims to determine whether LightGBM remains the best performing algorithm across different datasets, whether the reported accuracy can be reproduced, and whether any performance degradation occurs. If degradation is found, the study will further examine the possible causes before making any changes to the pipeline. The expected outcome is either (a) evidence that the pipeline can generalize well across independently collected Android malware datasets, or (b) a clear understanding of where and why its performance changes. The findings can then be used as a basis for further refinement of the pipeline.
C(ii)	**(a) Project background**

	Previous studies on Android malware detection including studies using the CICMalDroid2020 dataset, have reported good classification performance. However, most of these studies evaluate their methods using only a single dataset. This creates a research gap because it is still unclear whether the same detection pipeline, including its preprocessing methods such as normalization and oversampling, can perform well on a different but structurally similar dataset. Without cross-dataset evaluation, the performance of a pipeline may be specific to the dataset used rather than being generally applicable.
	(b) Objectives of the project
	This project has the following objectives: • To evaluate the reproducibility of a previously validated LightGBM-based detection pipeline (Log1p + RobustScaler + in-fold SMOTE) using an independent Android malware dataset, KronoDroid. • To compare the performance of LightGBM, XGBoost, RandomForest, and ExtraTrees on the KronoDroid dataset and determine whether LightGBM still performs well compared to the other algorithms. • To investigate the possible causes of any significant changes in performance between the CICMalDroid2020 and KronoDroid datasets. • To provide an evidence-based conclusion on whether the existing pipeline can be used across different Android malware datasets or needs further improvement.
	(c) Scope of the project
	• **Dataset:** KronoDroid real-device subnet, using its 289 dynamic system-call features and following the same general preprocessing approach as the previous study. • **Pipeline:** The same pipeline structure will be used, including deduplication, Log1p transformation, RobustScaler, in-fold SMOTE, and model comparison using LightGBM, XGBoost, RandomForest, and ExtraTrees. The evaluation will use an 80:20 hold-out and 5-Fold Cross Validation. • **Evaluation Metrics:** Accuracy, Presicion, Recall, and F1-Score. • **Out of Scope:** Adversarial robustness testing, concept-drift modelling, real time or on-device deployment, and deep learning methods. These topics may be considered as future work.
	(d) Expected Outcome / Proposed Solution
	The expected output is a comparative performance report between the two independently collected Android malware datasets. The report will include an analysis of the possible reasons for any differences in model performance and a conclusion about how well the existing pipeline generalizes across different datasets. This study is intended as a direct extension and validation of the researcher's previous work.
D	**REFERENCES**
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